Is Finishing a Skill?

Code

An expected-goals model built from 100,864 shots, benchmarked against StatsBomb's own, and used to separate finishing from luck

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Interactive application

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Summary

Expected goals (xG) is the probability that an average player scores a given chance. I built an xG model from every shot in StatsBomb's free event data (100,864 shots from 3,961 men's and women's matches) and used it to ask how much of "great finishing" is skill.

- **The model is competitive with a professional one.** On 811 held-out matches it scores a log loss of 0.2604 against 0.2640 for StatsBomb's own xG (ROC AUC 0.816 vs 0.809). Most of the gain is on women's matches.
- **Where the goalkeeper stands matters more than the shot angle.** The freeze frame (every player's position at the moment of the shot) carries most of the signal.
- **A men's model works for women's football.** Trained only on men's shots, it is well calibrated on women's matches.
- **For almost every player, beating xG is luck.** Over-performance in one half of a player's matches barely predicts the other half (r = 0.02, 146 players).
- **Messi is the exception.** Even measured against his Barcelona team-mates, who also beat xG, he scored 68 more non-penalty goals than expected, about 2.3 times the range luck allows.

1 Data

StatsBomb Open Data publishes full event data for selected competitions: every shot with its location, body part and technique, the pass that set it up, and a **freeze frame** (the position of

every visible player when the ball was struck). Penalty shoot-outs are excluded.

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Table 1: Shots by gender. The open data favours some teams and tournaments (Barcelona 2004–2021, World Cups, Euros), so it is not a random sample of football.

gender	matches	shots	goals	competitions	goals per shot
Women	1310	34286	3842	7	0.112
Men	2651	66578	7244	17	0.109

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Figure 1

2 Features and model

`features.shot_features()` builds every feature. The training script, the evaluation and the app's simulator all call it, so a shot placed in the app is scored exactly like a real one.

- **Geometry:** distance to goal, the angle the goal mouth subtends, how wide the shot is.
- **Freeze frame:** defenders and team-mates in the shooting cone; the nearest defender and how many are within 3 and 5 yards; the keeper's distance from goal and from the shooter, how far the keeper has come out, and how far he is off the line of the shot.
- **Context:** body part, technique, first-time, one-on-one, under pressure, phase of play, and the assist (cross, cut-back, through ball, its height and length).

The model is LightGBM, with the number of trees chosen by 5-fold cross-validation grouped by match (shots from one match never sit on both sides of a split). Penalties get a constant: the training conversion rate, 0.763.

2.1 Two fixes found by using the model

Building the simulator exposed two flaws that the test metrics could not see:

1. **Wide shots looked better than central ones.** The first model gave a shot from the touchline 25 yards out *more* xG than a central one at the same distance. About 35 freak goals from wide areas (mishit crosses) had pulled the trees. **Monotonic constraints** fixed it: xG must fall with distance, width and defenders in the cone, and rise with the angle.

2. **“Keeper far away” meant “easy goal”.** In real data a keeper 40 yards from the shooter is usually one who has left his goal, so the model rewarded large distances. That fired wrongly for any long or wide shot with the keeper on his line. The distance is now **capped at 12 yards**, with a separate “keeper advance” feature (how far he has come out towards the shooter) carrying the real signal.

Both changes cost almost nothing in held-out log loss (about +0.0003) and make the model behave sensibly everywhere on the pitch.

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Figure 2

3 How good is it?

Twenty percent of matches, chosen within every competition-season, were held out before any modelling. Table 2 scores three models on their non-penalty shots: the textbook distance-and-angle logistic regression, this project’s model, and **StatsBomb’s own xG**, which ships with the data.

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Table 2: Held-out non-penalty shots. Log loss and Brier: lower is better. ROC AUC: how well the model ranks goals above misses. Calibration error: shot-weighted gap between xG and the actual goal rate.

model	Log loss	Brier	ROC AUC	Calibration error	Goals	Total xG
Distance + angle (logistic)	0.2939	0.0836	0.739	0.0122	2105	2094.0
This project (LightGBM)	0.2604	0.0746	0.816	0.0045	2105	2111.0
StatsBomb xG	0.2640	0.0751	0.809	0.0109	2105	1984.0

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Figure 3

Is the comparison fair? Not entirely, and the biases run both ways. StatsBomb’s model is built on far more data than is public and may well have seen these very matches; but it was not tuned to these competitions, and this model was. Part of a log-loss edge can come from calibration alone

(StatsBomb's values add up to 1,984 goals where 2,105 were scored). So I re-calibrated StatsBomb's values on the training matches (Platt scaling, separately by gender) and compared again: this model is still ahead by **0.0033** in log loss (95% CI 0.0010 to 0.0058, bootstrapping the 811 test matches). ROC AUC, which ignores calibration altogether, tells the same story. The edge is real but small. The fair summary is *"on par with StatsBomb on men's football, slightly better on women's"*, not "better than StatsBomb".

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Figure 4

4 Men's and women's football

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Table 3: Held-out log loss by training data (rows) and test matches (columns). Lower is better.

test_on model	Men's matches	Women's matches
Trained on both	0.2560	0.2696
Trained on men's shots	0.2573	0.2725
Trained on women's shots	0.2584	0.2717

A model that has **never seen a women's match** predicts 707 goals on the women's test matches, where 702 were scored: the same chances go in at the same rates. Adding women's shots improves the model on *both* (it is simply more data), which is also why this project beats StatsBomb mostly on women's matches: StatsBomb's public model is presumably fitted mostly to men's football.

5 Is finishing a skill?

All numbers here use **out-of-fold xG**: every shot is scored by a model that never saw its match, so a player cannot "beat" a model fitted to their own shots.

5.1 Luck is large

Even an average finisher drifts above or below xG. Simulating each shot as a coin flip with probability equal to its xG gives the range luck alone produces for the *same* chances (Figure 5).

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Figure 5

5.2 Hot streaks don't repeat

If finishing were a strong skill, a player who beats xG in half their matches would beat it in the other half too. Splitting each player's matches into alternating halves (at least 40 shots in each), the correlation of per-shot goals-minus-xG between halves is **r = 0.02** (95% CI -0.16 to 0.18; 146 players): 0.02 for men, 0.00 for women. For a typical player, a season of hot finishing says almost nothing about the next. That doesn't mean finishing skill is zero, only that with a few dozen shots per half it is too small to tell apart from luck.

5.3 ...except Messi

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Table 4: Non-penalty goals vs out-of-fold xG. Luck: the 95% range of goals − xG for an average finisher on the same shots.

	group	shots	goals	xg	goals − xG	luck_low	luck_high
0	Lionel Andrés Messi Cuccittini (Barcelona)	2288	414	319.1	94.9	-28.1	29.9
1	Barcelona team-mates	5972	858	790.8	67.2	-41.8	45.2
2	Lionel Andrés Messi Cuccittini (other teams)	295	28	34.7	-6.7	-9.7	10.3
3	All other teams	91486	8961	9096.9	-135.9	-162.9	162.1

Messi scored **414** non-penalty goals for Barcelona from **319** xG. But his team-mates *also* beat xG, by 8%, more than luck allows. Part of the gap is therefore a team effect the model cannot see: the quality of chances Barcelona created, which a freeze frame only partly captures. Scaling Messi's xG by his team-mates' rate still leaves **+68 goals**, against a luck range of about ±29. Away from Barcelona (Argentina, PSG, Inter Miami; only 295 shots) his finishing is within luck.

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Figure 6

6 Limitations

- **Not a random sample.** The open data covers Barcelona's 2004–2021 seasons in full, several complete 2015/16 leagues, and many tournaments. Findings describe these matches.
- **What xG can't see.** Freeze frames give positions, not the ball's spin, the shooter's balance, or which is their stronger foot. Some of what looks like finishing skill is information the model lacks.
- **Team effects.** As Barcelona shows, a team can systematically create better chances than its freeze frames suggest. Player-level finishing should be judged against team-mates, not just the model.
- **The StatsBomb comparison** is informative, not a head-to-head contest: the two models were built for different purposes on different data.

7 Reproduce

```
python scripts/download_data.py   # StatsBomb Open Data → data/processed/shots.parquet
python scripts/train.py           # model, held-out metrics
python scripts/analyze.py         # out-of-fold xG, transfer, finishing analyses → results/
python scripts/build_app_data.py  # compact data for the app
quarto render report/report.qmd
```

Data: StatsBomb Open Data, used under its non-commercial licence with attribution.